

Probability & Distributions Notes

Structured from Handwritten Notes

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1 Fundamentals & Inequalities

1.1 Expected Value and Variance

- **Discrete:** $\mu = E(X) = \sum xf(x)$, and $\sigma^2 = Var(X) = \sum (x - \mu)^2 f(x)$.
- **Continuous:** $\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$, and $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$.
- **Properties:**
 - $E(aX + b) = aE(X) + b$
 - $Var(aX + b) = a^2 Var(X)$
 - $Var(X) = E(X^2) - (E(X))^2$
- **Moments:** The k -th moment is $E(X^k)$. The k -th central moment is $E((X - \mu)^k)$.

Important Inequalities

- **Markov's Inequality:** For $c > 0$, $P(|X| \geq c) \leq \frac{E(|X|)}{c}$.
- **Chebyshev's Inequality:** $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$.
- **Jensen's Inequality:** If $g(x)$ is a convex function ($g''(x) \geq 0$), then $g(E(X)) \leq E(g(X))$. Equality holds only if X is a constant or $g(x)$ is strictly linear.

2 Generating Functions

2.1 Probability Generating Function (PGF)

For a discrete random variable X :

$$G_X(t) = E(t^X) = \sum_{x=0}^{\infty} t^x P(X = x)$$

- $G_X(1) = 1$
- $E(X) = G'_X(1)$
- $Var(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2$

2.2 Moment Generating Function (MGF)

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \quad (\text{Continuous})$$

- $E(X^k) = M_X^{(k)}(0)$ (The k -th derivative evaluated at $t = 0$).
- $M_{aX+b}(t) = e^{bt} M_X(at)$.
- If X, Y are independent, $M_{X+Y}(t) = M_X(t) M_Y(t)$.

3 Discrete Distributions

Key Discrete Distributions

1. Geometric Distribution $X \sim GEO(p)$:

- X = number of trials needed to get the 1st success.
- $P(X = x) = pq^{x-1}$ for $x = 1, 2, \dots$ (where $q = 1 - p$).
- $E(X) = \frac{1}{p}$, $Var(X) = \frac{q}{p^2}$.
- **Memoryless Property:** $P(X > k + j | X > k) = P(X > j)$.

2. Negative Binomial Distribution $X \sim NEGBIN(r, p)$:

- X = number of trials needed for the r -th success.
- $P(X = x) = \binom{x-1}{r-1} p^r q^{x-r}$.
- $E(X) = \frac{r}{p}$, $Var(X) = \frac{rq}{p^2}$.

3. Poisson Distribution $X \sim POI(\mu)$:

- X = number of hits in a certain interval. $\mu = \lambda t$ (intensity).
- $P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$ for $x = 0, 1, 2, \dots$
- $E(X) = \mu$, $Var(X) = \mu$, $M_X(t) = e^{\mu(e^t - 1)}$.
- **Approximation:** $BIN(n, p) \approx POI(\mu)$ if $n \rightarrow \infty$, $p \leq 0.05$, and $np = \mu$.

4 Continuous Distributions

4.1 Uniform and Exponential

- **Uniform** $UNIF(a, b)$: $f(x) = \frac{1}{b-a}$. $E(X) = \frac{a+b}{2}$, $Var(X) = \frac{(b-a)^2}{12}$.
- **Exponential** $EXP(\lambda)$: $f(x) = \lambda e^{-\lambda x}$ for $x > 0$. $E(X) = \frac{1}{\lambda}$, $Var(X) = \frac{1}{\lambda^2}$.
 - Has the continuous memoryless property: $P(X > a + b | X > a) = P(X > b)$.
 - **Two-parameter Exp:** $X \sim Exp(\lambda, \eta)$. $f(x) = \lambda e^{-\lambda(x-\eta)}$. $E(X) = \eta + \frac{1}{\lambda}$.

4.2 Gamma, Chi-Square, and Weibull

- **Gamma** $GAM(\theta, r)$: Waiting time until the r -th event.

$$f(t) = \frac{1}{\theta^r \Gamma(r)} t^{r-1} e^{-t/\theta}$$

$$E(X) = r\theta, Var(X) = r\theta^2. \text{ (Note: } \Gamma(n) = (n-1)! \text{).}$$

- **Chi-Square** $\chi^2(v)$: A special case of Gamma where $\theta = 2, r = v/2$. $E(X) = v$, $Var(X) = 2v$. If $Z \sim N(0, 1)$, then $Z^2 \sim \chi^2(1)$.
- **Weibull** $WEI(\theta, \beta)$: $f(x) = \frac{\beta x^{\beta-1}}{\theta^\beta} e^{-(x/\theta)^\beta}$.

4.3 Normal Distribution

- **Normal** $N(\mu, \sigma^2)$: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$.
- **Standardization**: If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.
- **Central Limit Theorem (CLT)**: For sample sum $S_n \sim N(n\mu, n\sigma^2)$ and sample mean $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ as $n \geq 30$.

5 Transformations & Reliability

Probability Integral Transformation (PIT)

If X is a continuous random variable with CDF $F(x)$, then the random variable defined by $Y = F(X)$ has a standard Uniform distribution: $Y \sim UNIF(0, 1)$.

- **Reliability function**: $R(t) = P(T > t) = 1 - F(t)$.
- **Hazard rate**: The conditional density of failure at time t given survival up to time t :

$$h(t) = \frac{f(t)}{R(t)} = \frac{f(t)}{1 - F(t)} = -\frac{d}{dt} \ln R(t)$$